15.IPV4 Addressing:

1. **Convert the following IP address to its 32-bit binary representation: 172.16.5.10. Write out each octet in binary, separated by dots.**

ANS :

P address: 172.16.5.10

Convert each octet to 8-bit binary:

- 172 → 10101100
- 16 → 00010000
- 5 → 00000101
- 10 → 00001010

Final Answer

172.16.5.10 = 10101100.00010000.00000101.00001010

2. **Given the binary IP address 11000000.10101000.00000001.00001111, convert it back to standard dotted decimal notation.**

   ANS:

   Convert each 8-bit octet from binary to decimal:

- 11000000 → 192
- 10101000 → 168
- 00000001 → 1
- 00001111 → 15

  Final Answer

  11000000.10101000.00000001.00001111 = 192.168.1.15

3. **You are given a list of 10 IP addresses used by different apps (e.g., 192.168.43.1, 8.8.8.8, 10.0.5.25, 172.20.14.2, 203.0.113.5, 172.15.255.255, 192.0.2.10, 192.168.1.100, 123.45.67.89, 172.16.0.1). For each, identify if it is a private or public IP address and explain your reasoning.**

   ANS:

   Private vs Public IP Addresses
   Private IPv4 addresses are defined in these three ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

IP Address	Type	Reason
192.168.43.1	Private	Falls within 192.168.0.0/16
8.8.8.8	Public	Not within any private range
10.0.5.25	Private	Falls within 10.0.0.0/8
172.20.14.2	Private	Falls within 172.16.0.0/12
203.0.113.5	Public*	Not within the three private ranges
172.15.255.255	Public	172.15.x.x is outside the private 172.16–172.31 range
192.0.2.10	Public*	Not within the private 192.168.x.x range
192.168.1.100	Private	Falls within 192.168.0.0/16
123.45.67.89	Public	Not within any private range
172.16.0.1	Private	Falls within 172.16.0.0/12

   Final classification

   Private:
   192.168.43.1, 10.0.5.25, 172.20.14.2, 192.168.1.100, 172.16.0.1
   Public-range addresses:
   8.8.8.8, 203.0.113.5, 172.15.255.255, 192.0.2.10, 123.45.67.89

**4. For the CIDR notations /20, /25, and /30, calculate the total number of usable IP addresses in each subnet. Show your working for each calculation.

Hint: Usable addresses = (2^{number of host bits}) minus 2 (for network and broadcast addresses).**

ANS:

CIDR Subnet Usable IP Addresses

Formula:

Usable IP addresses = 2^(host bits) – 2

For IPv4, there are 32 total bits.

CIDR	Host Bits	Calculation	Usable IP Addresses
/20	32 − 20 = 12	$2^{12} - 2 = 4096 - 2$	4,094
/25	32 − 25 = 7	$2^{7} - 2 = 128 - 2$	126
/30	32 − 30 = 2	$2^{2} - 2 = 4 - 2$	2

Final Answer

- /20 → 4,094 usable IP addresses
- /25 → 126 usable IP addresses
- /30 → 2 usable IP addresses

5.Imagine you are setting up a small LAN for a gaming night with friends using your WiFi router at home. Assign static private IP addresses to 5 devices (laptop, phone, tablet, smart TV, and PlayStation) using the 192.168.0.0/24 range. List the IP you would assign to each device and explain why you chose that range.

Static IP Address Assignment for Gaming LAN

For the gaming night, I would use the 192.168.0.0/24 private network. A /24 subnet provides addresses from 192.168.0.0 to 192.168.0.255. The network address (.0) and broadcast address (.255) are reserved, so I would assign devices addresses in between.

Device	Static IP Address
💻 Laptop	192.168.0.10
📱 Phone	192.168.0.11
📱 Tablet	192.168.0.12
📺 Smart TV	192.168.0.13
🎮 PlayStation	192.168.0.14

Why choose 192.168.0.0/24?

- Private IP range: 192.168.0.0/24 is part of the private IPv4 address space.
- Suitable for a home LAN: Devices can communicate with each other through the Wi-Fi router.
- Easy to remember: Assigning .10–.14 makes the devices easy to identify.
- Enough addresses: A /24 network has 256 total addresses, with 254 normally usable host addresses.
- No direct Internet exposure: These private addresses are used inside the home network; the router handles Internet access using NAT.

Example: The router could be 192.168.0.1, while the five gaming-night devices use .10 through .14.

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